Network Traffic Verification

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PCAP:
https://github.com/asamc/TutorAndQuizBot/blob/main/attachments/WireShark_TrafficCapture.pcapng

Purpose

The purpose of this capture is to monitor the network traffic while the Quiz Agent is in use. The agent reads the lecture slides, sends the slide text to a local Llama model, and receives quiz questions back.

Our goal is to confirm that none of the course documents leave the local machine at any point during this process. We used Wireshark to monitor all traffic inside the computer and on the Wi-Fi connection to verify this.

Setup

- **Tool**: Wireshark with Npcap, with loopback capture turned on.
- **Interfaces captured**: the loopback adapter (traffic inside the computer) and Wi-Fi (traffic leaving the computer).
- **Model**: Llama 3.2 (3B) running in Ollama at 127.0.0.1:11434.
- **Documents**: the course lecture slides, including the OSI Security Architecture deck.
- **Computer's** Wi-Fi address: 192.168.1.249.

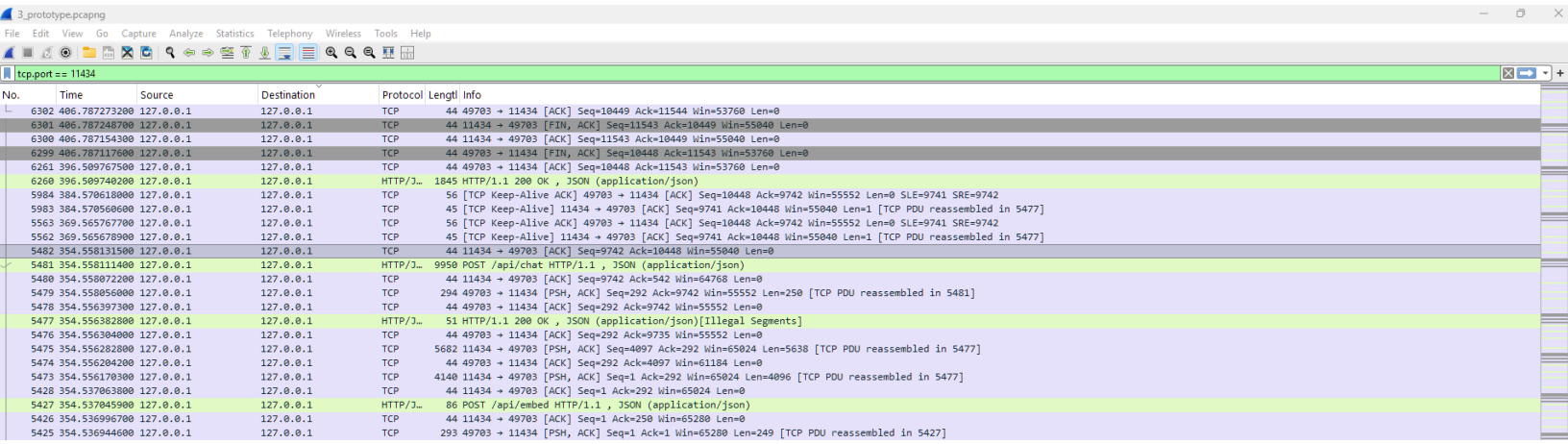
Steps

1. Started the quiz app with uv run python -m ui.terminal. Waited at the main menu before capturing, so the app's startup traffic would not be included.
2. Opened Wireshark and selected two interfaces: **Adapter for loopback traffic capture**, which shows traffic inside the computer, and **Wi-Fi**, which shows traffic leaving the computer. We then started the capture.
3. Took a quiz. We answered each question and noted the source slide shown after each answer, so we knew which slide and keyword text to search for later.
4. Stopped the capture once the quiz was finished, then saved it in pcap format.
5. Applied the filters in the next section to check where the quiz traffic went and whether any slide text appeared in traffic leaving the computer. We took screenshots of each result as evidence.

Results & Analysis

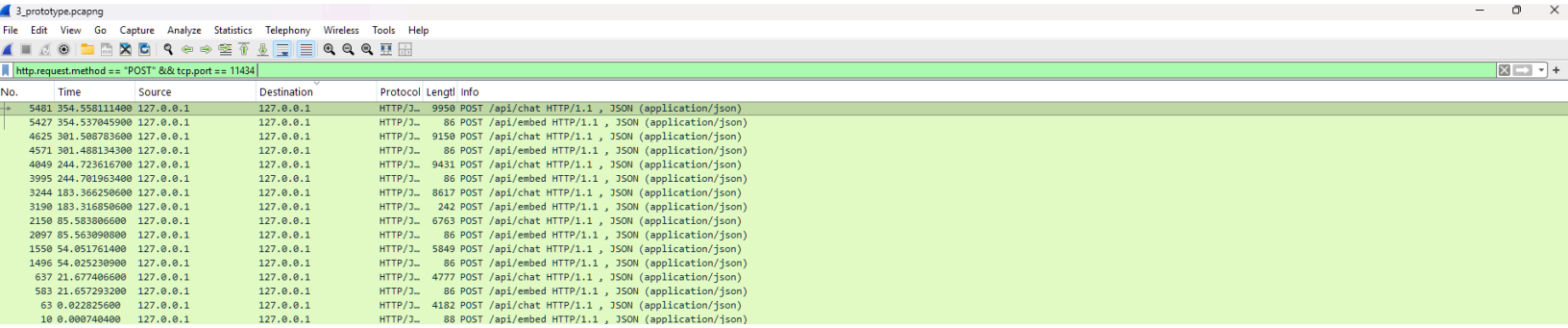
Filter	What it shows	Expected Result	Result
`tcp.port == 11434`	Quiz ↔ local model traffic	Only 127.0.0.1	✓
`http.request.method == "POST" && tcp.port == 11434`	Each prompt sent to the model	1 per question	✓
`frame matches "(?i)OSI Security Architecture"`	Where the slide text appears	Loopback only	✓
`frame matches "(?i)OSI Security Architecture" && !(ip.dst == 127.0.0.1)`	Slide text sent off the machine	0 packets	✓
`tcp.port == 11434 && !(ip.addr == 127.0.0.1)`	Outside access to the model	0 packets	✓
`dns`	Websites looked up	Background only	✓

1. `tcp.port == 11434`



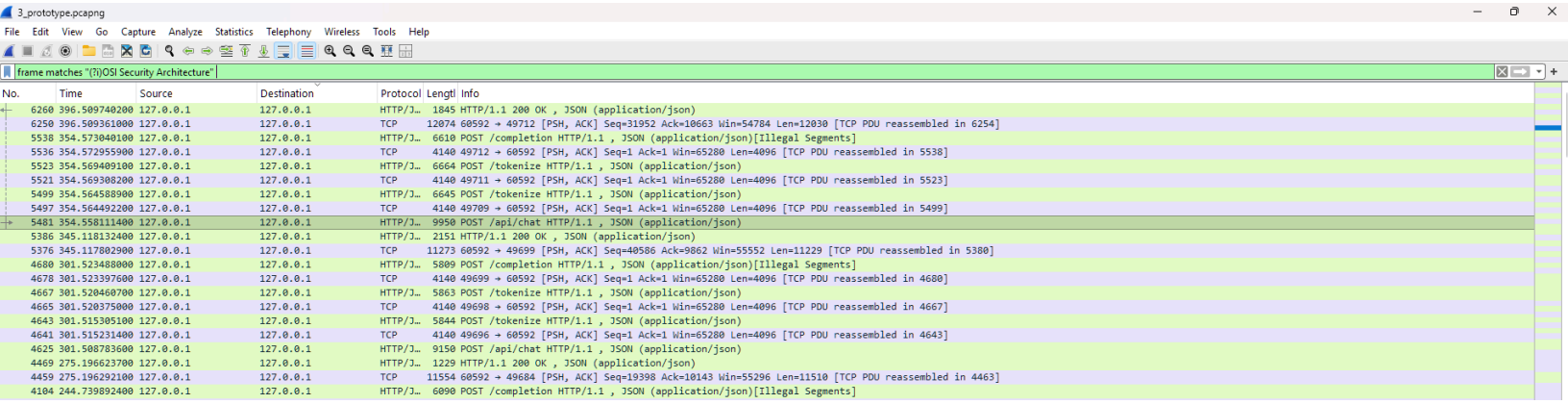
All traffic between the quiz app and the model went between 127.0.0.1 and 127.0.0.1. This means the quiz only talks to the model on the same computer.

2. `http.request.method == "POST" && tcp.port == 11434`



This filter shows each request the quiz app sent to the local model, one for every question it generated. All of these requests went to 127.0.0.1, so every prompt stayed on the computer.

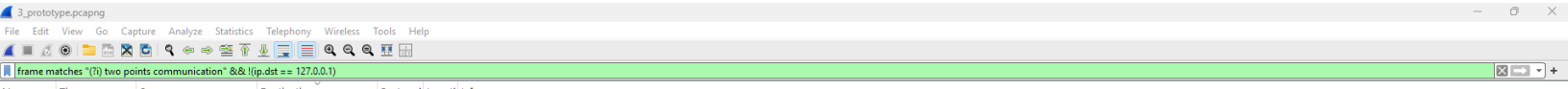
3. `frame matches "(?i)OSI Security Architecture"`



The slide text was found in the capture, but only in loopback traffic. This shows the slide content was sent to the model, and that it stayed inside the computer.

We also tested with other phrases from the slides (e.g., "Hyperledger", "two points communication"). Each appeared only in loopback traffic to 127.0.0.1.

```
4. frame matches "(?i)OSI Security Architecture" && !(ip.dst ==
127.0.0.1)
```



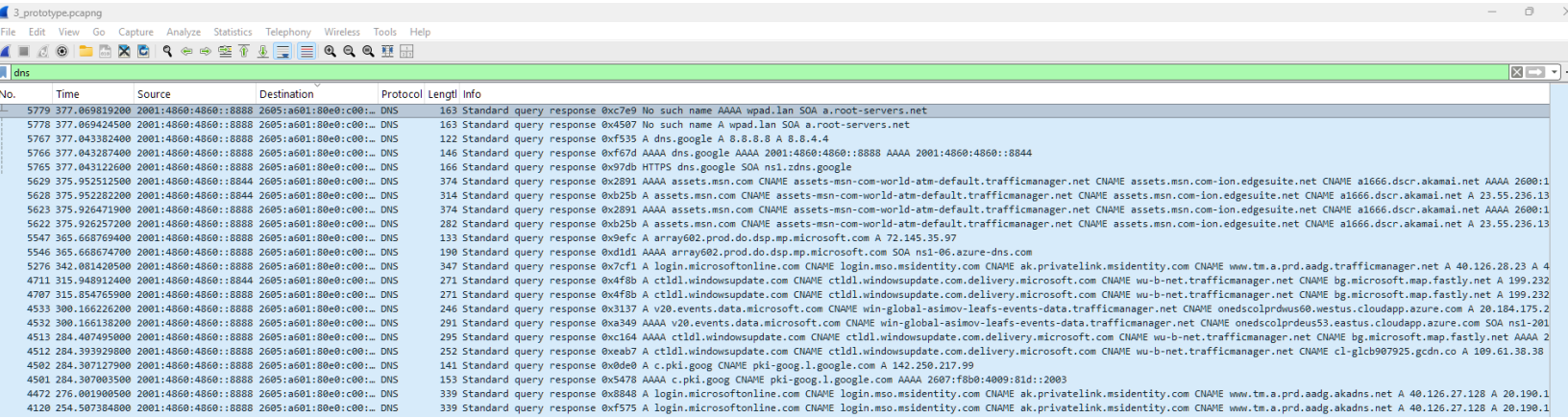
This search returned 0 packets. This means no slide text was sent out from the computer over Wi-Fi.

```
5. tcp.port == 11434 && !(ip.addr == 127.0.0.1)
```



This search returned 0 packets. This means no device outside the computer connected to the model.

```
6. dns
```



The only website lookups came from background apps like Windows and VS Code. None of them came from the quiz app, which means the quiz did not try to reach the internet.

Conclusion

The capture shows that the Quiz Agent keeps all course content on the local computer. Every request to the model went to 127.0.0.1, and no slide text appeared in any traffic sent out over Wi-Fi. The other traffic in the capture came from background apps, not from the quiz.

The capture supports our privacy goal: the course documents never left the machine while the quiz was running.